Scikit-Learn

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Python Libraries for Data Science

NumPy:

- introduces objects for multidimensional arrays and matrices, as well as functions that allow to easily perform advanced mathematical and statistical operations on those objects
- provides vectorization of mathematical operations on arrays and matrices which significantly improves the performance
- many other python libraries are built on NumPy

Link: http://www.numpy.org/

Python Libraries for Data Science

SciPy: (Scientific Python)

- collection of algorithms for linear algebra, differential equations, numerical integration, optimization, statistics and more
- part of SciPy Stack
- built on NumPy

Link: https://www.scipy.org/scipylib/

Python Libraries for Data Science

SciKit-Learn

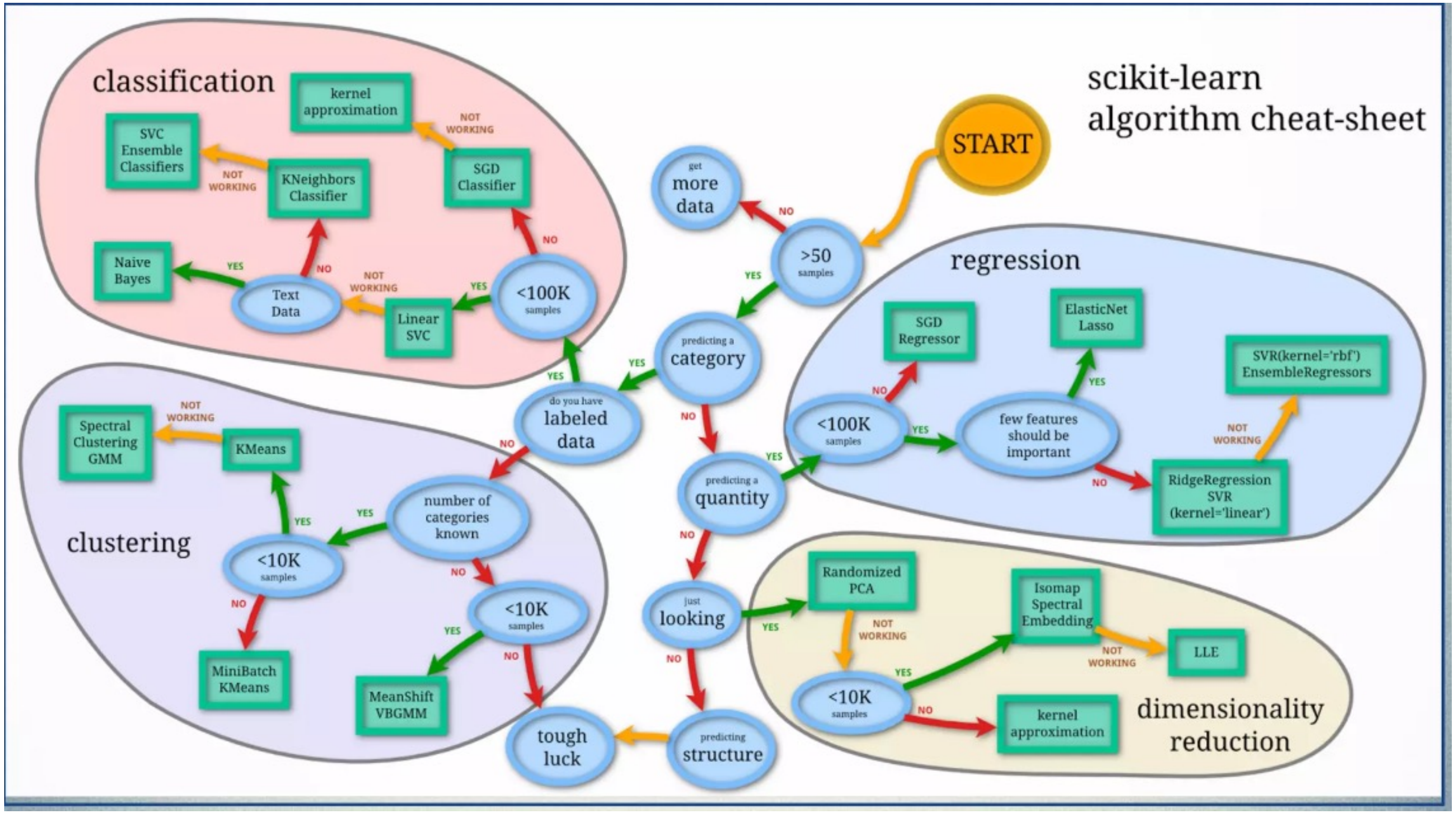
- Extensions to SciPy, called SciKits (Short form: sklearn)
- provides machine learning algorithms: classification, regression, clustering, model validation etc.
- built on NumPy, SciPy and matplotlib
- Sits on top of C libraries: LAPACK, LibSVM, and Cython
- Open source: BSD License (part of Linux)
- Best general ML framework

Link: http://scikit-learn.org/

Primary Features

- Generalized Linear Models
- SVMs, kNN, Bayes, Decision Trees, Ensembles
- Clustering and Density algorithms
- Cross Validation
- Grid Search
- Pipelining
- Model Evaluations
- Dataset Transformations
- Dataset Loading

Scikit-Learn CheatSheet



Scikit-Learn API

O*bject-oriented* interface centered around the concept of an *Estimator*:

> "An estimator is any object that learns from data; it may be a classification, regression or clustering algorithm or a transformer that extracts/filters useful features from raw data."
>
> - Scikit-Learn Tutorial

- Model = EstimatorObject()
- Model.fit(dataset.data, dataset.target)
  - dataset.data = dataset
  - dataset.target = labels
- Model.predict(dataset.data)

Estimators

```python
class Estimator(object):

    def fit(self, X, y=None):
        """Fits estimator to data. """
        # set state of ``self``
        return self

    def predict(self, X):
        """Predict response of ``X``. """
        # compute predictions ``pred``
        return pred
```

The Scikit-Learn Estimator API

Estimator Example

Estimators

- `fit(X,y)` sets the state of the estimator.
- `X` is usually a 2D numpy array of shape `(num_samples, num_features)`.
- `y` is a 1D array with shape `(n_samples,)`
- `predict(X)` returns the class or value
- `predict_proba()` returns a 2D array of shape `(n_samples, n_classes)`

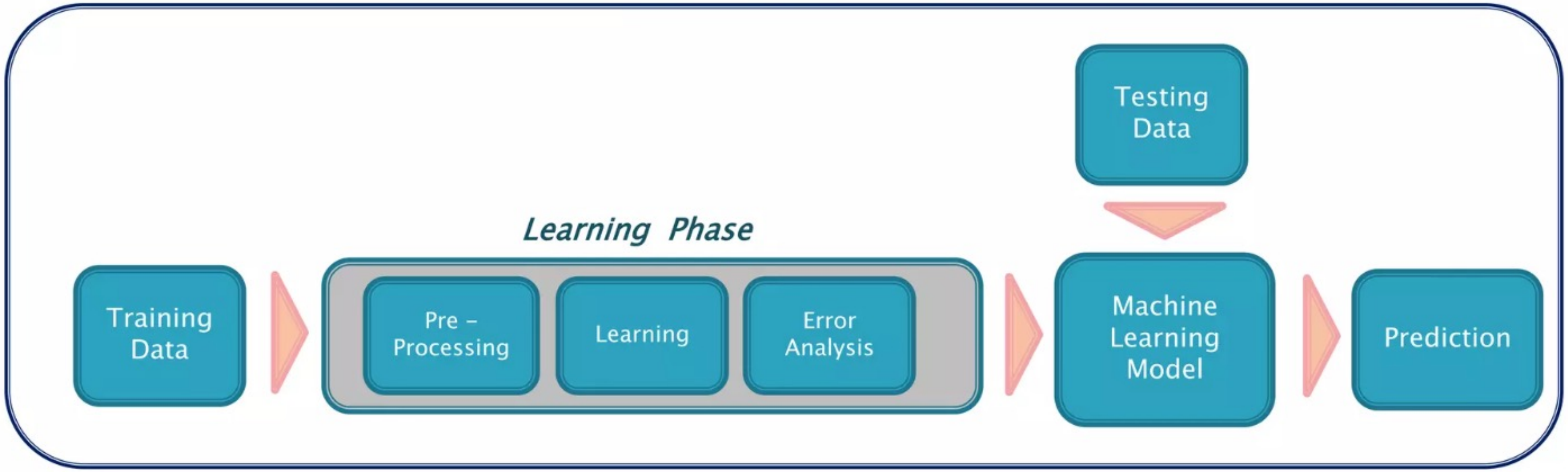
SVM Estimator

```python
from sklearn import svm

estimator = svm.SVC(gamma=0.001)
estimator.fit(X, y)
estimator.predict(x)
```

ML workflow

Machine learning algorithms are computer system that can adapt and learn from their experience



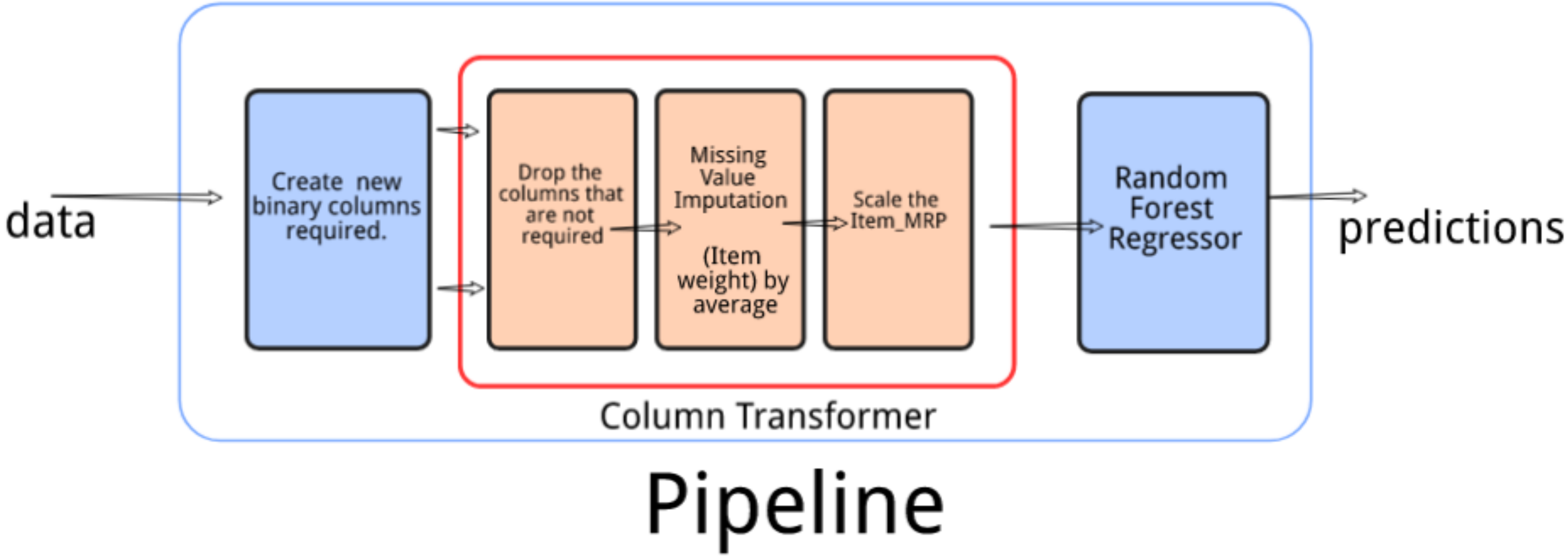
Two of the most widely adopted machine learning methods are

- Supervised learning are trained using labeled examples, such as an input where the desired output is known e.g. regression or classification
- Unsupervised learning is used against data that has no historical labels e.g. cluster analysis
- Third ML paradigm is Semi-supervised learning which is used when there are strong reasons to believe that a typical pattern exists in data such that the given pattern can be quantified via models.

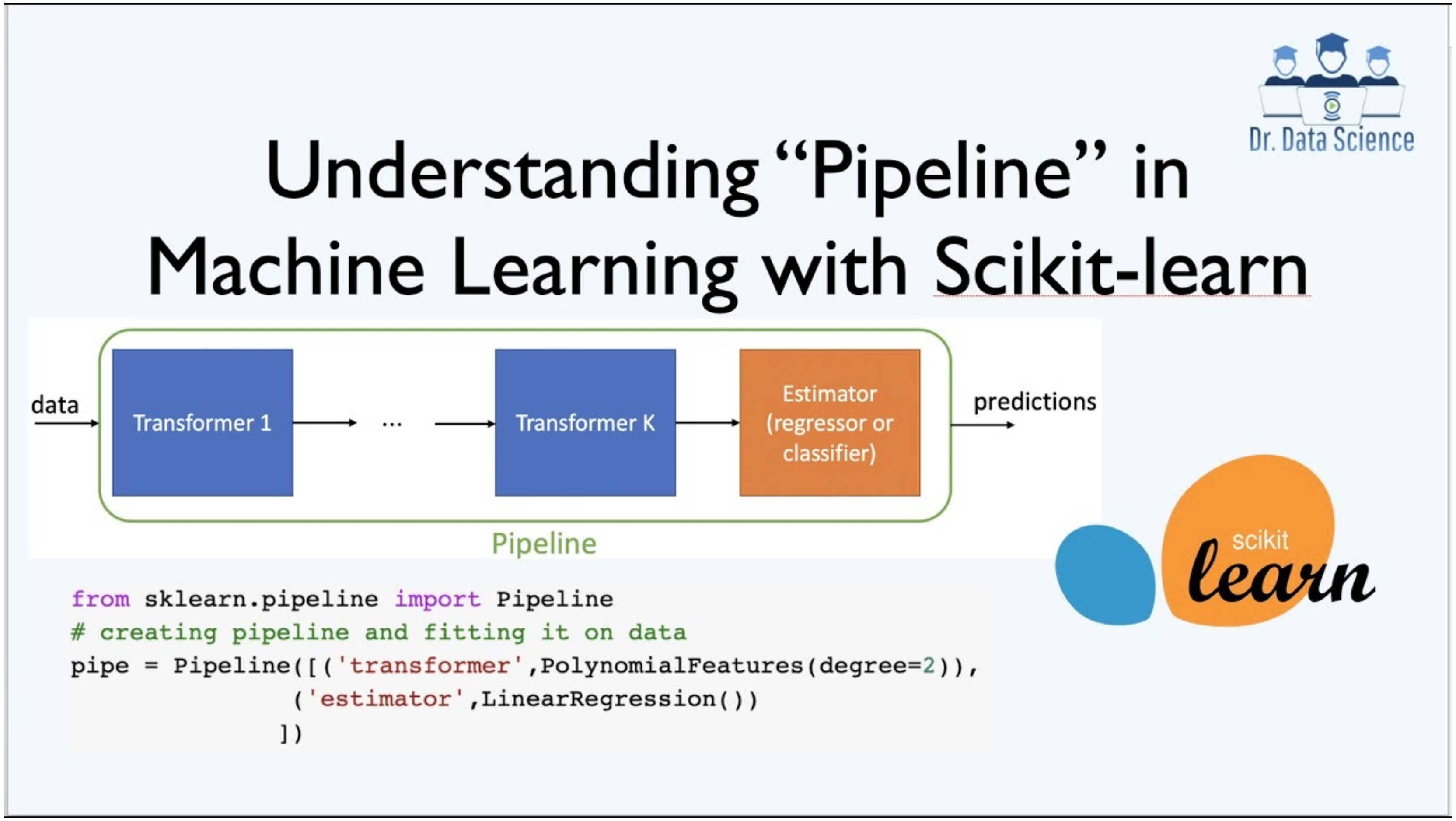
Pipeline

pipeline.fit() when using the training data

pipeline.predict() on test data



Pipeline


Dr. Data Science
Understanding "Pipeline" in
Machine Learning with Scikit-learn
data
Transformer 1
...
Transformer K
Estimator (regressor or classifier)
predictions
Pipeline
from sklearn.pipeline import Pipeline
# creating pipeline and fitting it on data
pipe = Pipeline([('transformer',PolynomialFeatures(degree=2)),
('estimator',LinearRegression())
])
scikit
learn

Training Validation & Testing

Statistical Modeling Methodology

- Training data used to train the model
- Testing data used to test the accuracy of the model

Machine Learning Modeling Methodology

- Train data used to train model by pairing input with expected output
- Validation data used to check how well model has been trained and to estimate model properties (mean error, classification error, precision, recall etc.)
- Finally calculate accuracy on Test data

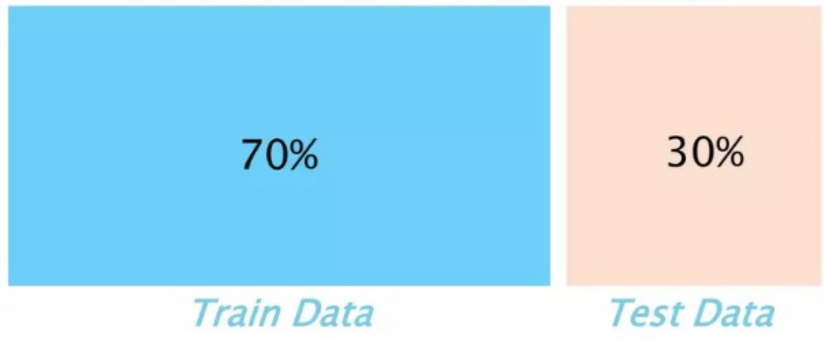
➤ ***In first part you look at your model and select the best performing approach using the validation data***

➤ ***Then you estimate the accuracy of the model approach based on test data***

Why separate Validation & Test data sets required ?

The error rate estimate of the final model on validation data will be biased (smaller than the true error rate) since the validation set is used to select the final model After assessing the final model on the test set, YOU MUST NOT tune the model any further!

Statistical Modeling Methodology



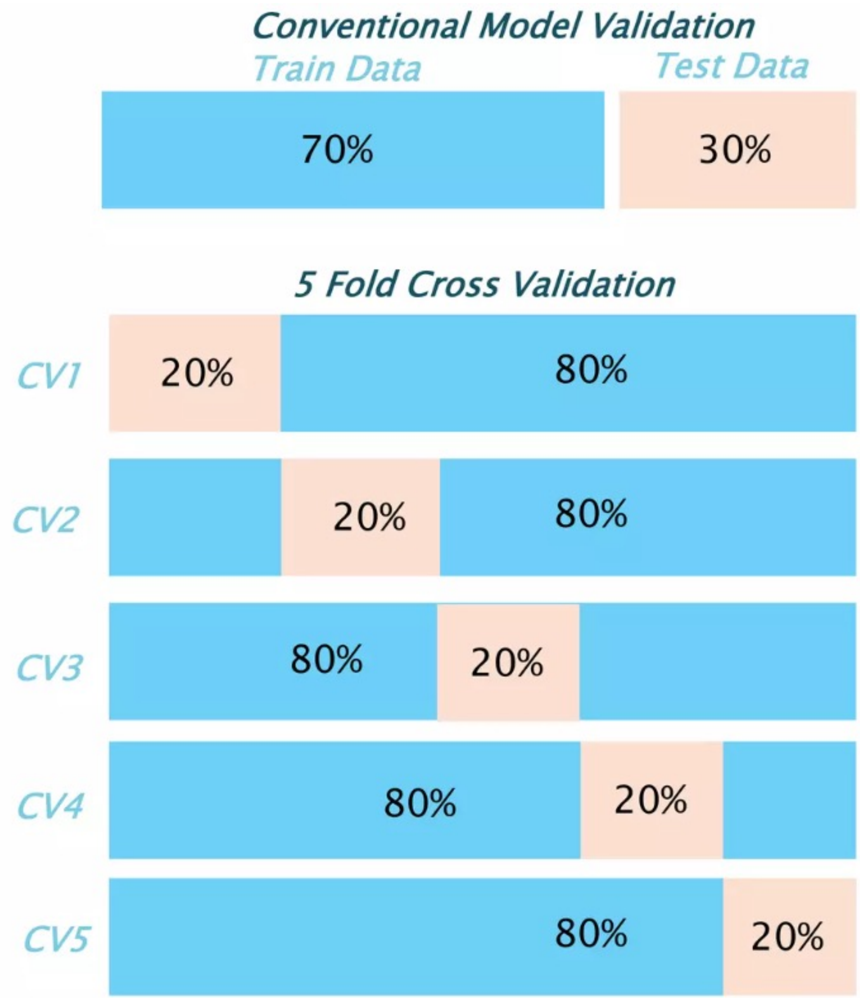
Machine Learning Modeling Methodology

Cross Validation

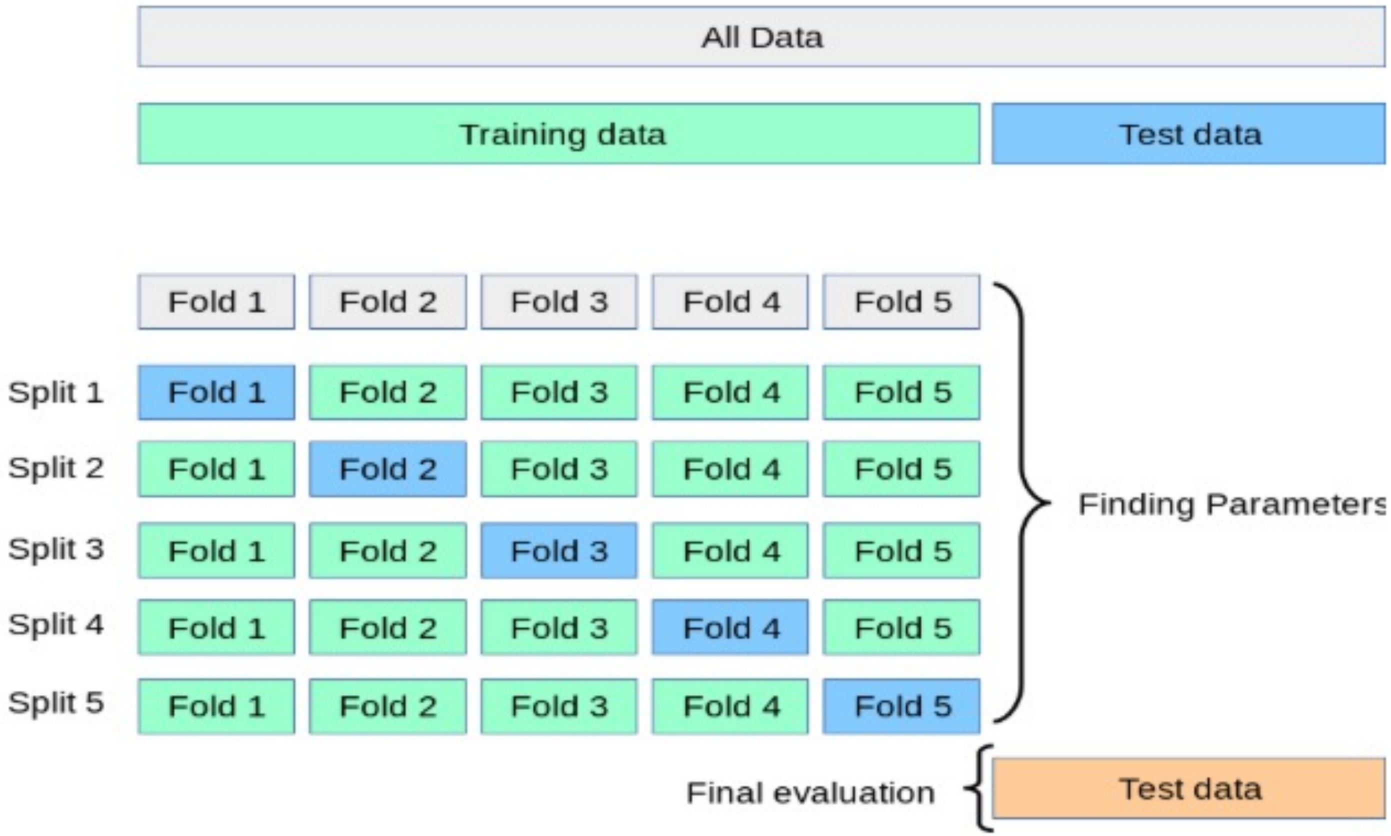
Cross Validation

- Cross validation improves the robustness of the models & provides the mean errors by averaging all possibilities as the models covers entire data points within with mix & match

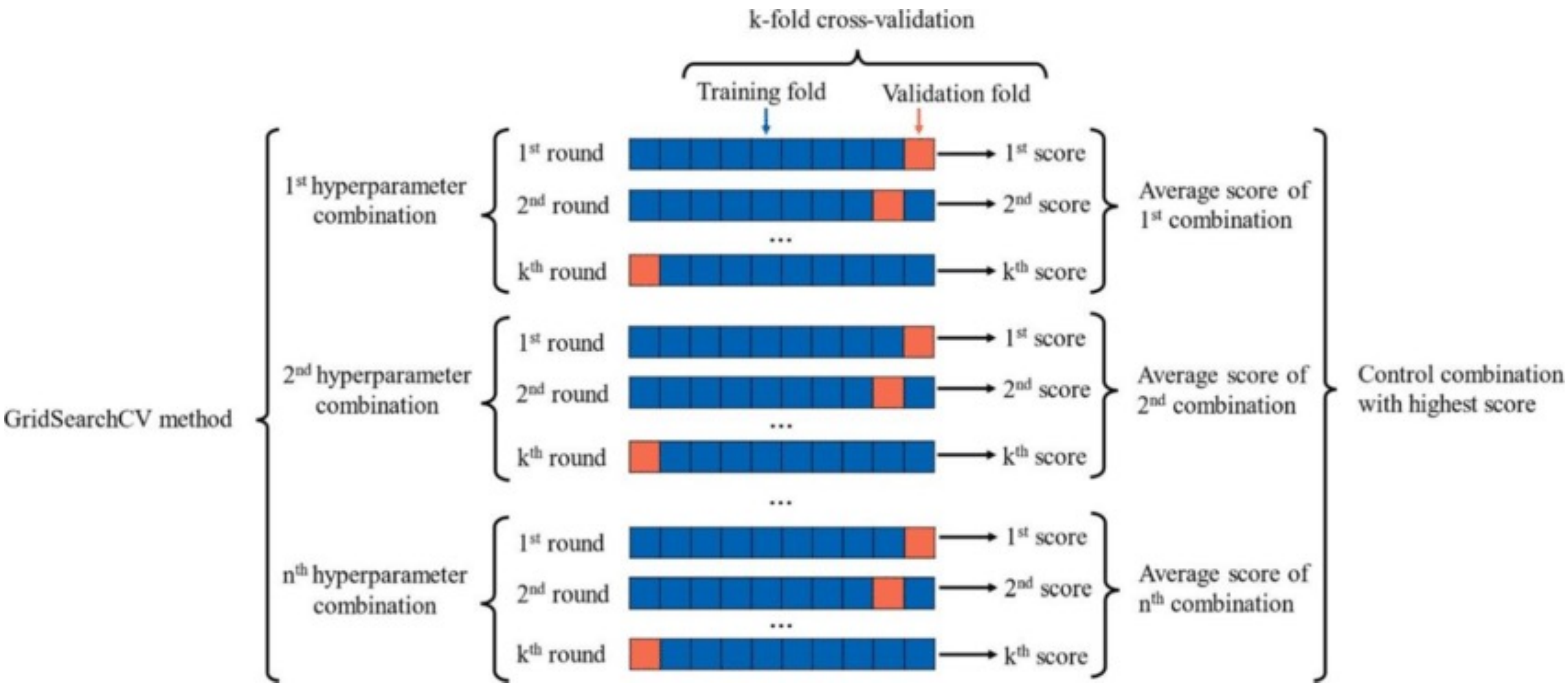
5 Fold CV Error
= (CV1 Error + CV2 Error+ CV3 Error+ CV4 Error+ CV5 Error) / 5



Cross Validation



GridSearchCV



When to Stop Training Model

Machine Learning Model Tuning

- Always keep a tab on train & validation errors while tuning the algorithm
- Stop increasing flexibility/degrees of the model when validation error starts increasing

Tuning of Machine Learning Models

ML Algorithms in Scikit-Learn

Supervised Learning

Classification/Regression

- Linear Regression
- Polynomial Regression
- Logistic Regression
- Decision Trees
- Random Forest
- Boosting
- Support Vector Machines
- KNN (K–Nearest Neighbors)
- Neural Networks
- Naïve Bayes

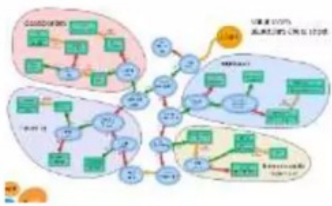
Unsupervised Learning

Clustering & Variable reduction

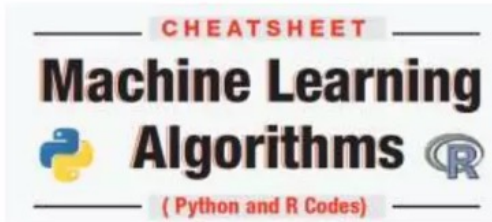
- K–means clustering
- PCA (Principal Component Analysis)

Supporting Techniques

- Cross Validation
- Gradient Descent
- Grid Search



ML Cheatsheet



ML Algorithms & Codes

Referecens

- 1. https://www.slideshare.net/PratapDangeti/machine-learning-with-scikitlearn-72720571
- 2. https://www.slideshare.net/SarahGuido/a-beginners-guide-to-machine-learning-with-scikitlearn
- 3. https://slideplayer.com/slide/15760442/
- 4. https://www.slideshare.net/BenjaminBengfort/introduction-to-machine-learning-with-scikitlearn